

Onyx — NetKit reference

Onyx · multi-process OS on Raspberry Pi 4

The reference of **NetKit** (`SD:/lib/netkit.so`, `user/Kits/netkit`): what it is for, how a program uses it, and every operation it exposes. The operations' part is made from the kit's headers by `tools/docgen/kitdocs.py` — the headers are the source. Overview of all the kits: [The Kits](#).

Contents

1. What it is
2. Using it
3. Index
4. `netkit/httpc.h`
5. `netkit/ftpfs.h`
6. `netkit/http.hpp`

What it is

NetKit is the network for the programs: a small HTTP client, the FTP volumes, and the HTTP/1.1 class with its TLS transport (`netkit/http.hpp`, included apart: still a header with its code). The sockets themselves are AppKit's (`kapi_tcp_*`).

Include	<code>#include "netkit/netkit.h"</code>
Link	<code>lib/netkit.imp.a</code> (C++) or <code>lib/netkit.imp_c.a</code> (C)
Library	<code>SD:/lib/netkit.so</code> — 21 entries in its table (<code>user/Kits/netkit/netkit.abi</code> , append-only)
Sources	<code>user/Kits/netkit/</code>

Using it

A page fetched (HTTP/1.0, no allocation: the caller gives the buffer):

```
#include "appkit/appkit.h"
#include "netkit/netkit.h"

static char buf[32 * 1024];

int main (void)
{
    http_response r;
    if (http_get ("http://example.com/", buf, sizeof buf, &r) < 0 || !r.ok)
    {
        ax_putln ("request failed");
        return 1;
    }
    kapi_stdout_write (r.body, (unsigned) r.body_len);
}
```

```
    return 0;
}
```

A web API (HTTP/1.1, headers, HTTPS with the TLS transport) — the `HttpClient` class:

```
#include "netkit/http.hpp"

static char buf[64 * 1024];
HttpClient http;
http.user_agent ("MyApp/1.0").accept ("application/json");
HttpResponse r = http.get ("http://api.example.com/status", buf, sizeof buf);
if (!r.is_error ()) { /* r.status, the body in buf */ }
```

An FTP server as a volume: once logged in, `FTP:host/dir/file` is a path like any other, for every program.

```
#include "netkit/netkit.h"

if (ftpfs_login ("ftp.example.com", "me", "secret", 1)) // 1: remembered
{
    void *f = kapi_open ("FTP:ftp.example.com/pub/readme.txt");
    /* kapi_read ... kapi_close */
}
```

The raw sockets (`kapi_tcp_connect`, `kapi_tcp_send...`) are AppKit's; NetKit is where protocols built on them belong.

Index

Everything the headers declare, in their order — the details are in each header's part below.

Name	What it does	Header
<code>http_request</code>	Perform an HTTP request.	<code>httpc.h</code>
<code>http_get</code>	Perform an HTTP request.	<code>httpc.h</code>
<code>http_post</code>	Perform an HTTP request.	<code>httpc.h</code>
<code>ftpfs_site</code>	(a type)	<code>ftpfs.h</code>
<code>ftpfs_obf_hex</code>	a password → its hexpass (XOR a fixed key, in hex: obfuscated, not encrypted) in out (cap bytes, cut to fit)	<code>ftpfs.h</code>
<code>ftpfs_unobf_hex</code>	a hexpass (<code>ftpfs_obf_hex</code>) → the password in out (cap bytes, cut to fit)	<code>ftpfs.h</code>
<code>ftpfs_parse_site</code>	Parse one file line	<code>ftpfs.h</code>
<code>ftpfs_format_site</code>	Format one file line (with '\n')	<code>ftpfs.h</code>
<code>ftpfs_text_fix</code>	A text file saved by a Windows editor	<code>ftpfs.h</code>

Name	What it does	Header
<code>ftpfs_load_sites</code>	The remembered servers, in file order.	<code>ftpfs.h</code>
<code>ftpfs_login</code>	The remembered servers, in file order.	<code>ftpfs.h</code>
<code>ftpfs_login_site</code>	The remembered servers, in file order.	<code>ftpfs.h</code>
<code>ftpfs_forget</code>	The remembered servers, in file order.	<code>ftpfs.h</code>
<code>HttpError</code>	Transport / protocol errors are reported as a NEGATIVE <code>HttpResponse::status</code> .	<code>http.hpp</code>
<code>slen</code>	The length of the string <code>s</code> (0 for a null pointer).	<code>http.hpp</code>
<code>lc</code>	<code>c</code> in lower case ('A'..'Z' only).	<code>http.hpp</code>
<code>cat</code>	Append <code>s</code> (or <code>n</code> bytes of <code>s</code>) to <code>d[*o]</code> , never overflowing cap	<code>http.hpp</code>
<code>catn</code>	Append <code>s</code> (or <code>n</code> bytes of <code>s</code>) to <code>d[*o]</code> , never overflowing cap	<code>http.hpp</code>
<code>cati</code>	Append <code>s</code> (or <code>n</code> bytes of <code>s</code>) to <code>d[*o]</code> , never overflowing cap	<code>http.hpp</code>
<code>find</code>	Case-insensitive lookup of a header in <code>[hdr, hdr+hlen)</code> .	<code>http.hpp</code>
<code>parse_int</code>	The decimal number at the start of <code>[s, s+n)</code> (its leading digits, no sign) → its value, 0 if there is no digit.	<code>http.hpp</code>
<code>contains_ci</code>	Case-insensitive	<code>http.hpp</code>
<code>dechunk</code>	Decode HTTP chunked transfer-coding in place over <code>[b, b+len)</code> .	<code>http.hpp</code>
<code>starts_ci</code>	Does the string <code>s</code> start with <code>pfx</code> , the case ignored?	<code>http.hpp</code>
<code>parse_url</code>	Parse "[http(s)://]host[:port] [/path]".	<code>http.hpp</code>
<code>resolve_redirect</code>	Resolve a redirect Location against the current absolute URL	<code>http.hpp</code>

Name	What it does	Header
	<code>base</code> , into <code>out[cap]</code> .	
Transport	A thin send/recv/close seam so HttpClient is oblivious to http vs https.	<code>http.hpp</code>
<code>tp_open</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
<code>tp_send</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
<code>tp_recv</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
<code>kapi_tcp_recv</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
<code>tp_close</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
<code>kapi_tcp_close</code>	Open the connection (and TLS handshake if requested).	<code>http.hpp</code>
HttpResponse	A parsed HTTP response.	<code>http.hpp</code>
HttpClient	Reusable HTTP/1.1 client.	<code>http.hpp</code>

netkit/httpc.h

httpc.h -- a minimal HTTP/1.0 client for Onyx apps, layered on the ABI v21 TCP socket calls (`kapi_tcp_connect/send/recv/close` + `kapi_net_status`).

Design choices for the Onyx user model:

- * Header-only (static inline) -- just #include it.
- * NO dynamic allocation (there is no user malloc): the CALLER provides the response buffer. Everything else lives on the app stack. So all memory is in the app's own address space (static .bss @ 8 GB / stack @ 16 GB); the TCP kapis copy between that buffer and the kernel socket buffers.
- * HTTP/1.0 + "Connection: close": the server closes the socket when done, so we just read to EOF -- no chunked-transfer decoding needed (1.0 never chunks).
- * Plain HTTP only (port 80 default) -- there is no TLS in the stack yet.
- * Blocking: polls the non-blocking `kapi_tcp_recv` with a timeout; fine for the occasional fetch (it briefly stalls the calling app, like an IRC connect).

typedef struct

```
{
    int    status;    // HTTP status code (200, 404, ...), 0 if unparsed
    int    ok;        // 1 if status is 2xx
    char *body;       // points INTO the caller's buffer (start of the body)
```

```

    int body_len;      // number of body bytes in the buffer
    int total_len;     // total bytes received (headers + body)
    int truncated;     // 1 if the response did not fit in the buffer
} http_response;

```

internals (hc_*)

Append a non-negative integer in decimal.

Parse "[http://]host[:port][/path]" into host / port / path (path defaults "/").

public API

Perform an HTTP request. method = "GET"/"POST"/...; xheaders = extra header lines ("Key: Value\r\n" each) or 0; body/body_len = request body or 0. The response (headers + body) is read into buf[cap]; resp (if non-0) is filled in, with resp→body pointing into buf. Returns the body length (>=0), or <0 on error (-1 bad URL, -2 connect failed).

```

int http_request (const char *method, const char *url,
                  const char *xheaders, const char *body, unsigned body_len,
                  char *buf, unsigned cap, http_response *resp);
int http_get (const char *url, char *buf, unsigned cap, http_response *r);
int http_post (const char *url, const char *body, unsigned len,
               char *buf, unsigned cap, http_response *r);

```

netkit/ftpfs.h

ftpfs.h -- talk to /bin/ftpfs (the FTP: / FTPS: file-system provider) and read its list of remembered servers. Header-only, no libc (C and C++, freestanding apps and newlib tools).

```

ftpfs_login ("ftp.example.com", "me", "secret", remember) -> 1 if delivered
hand a login over IPC (starting ftpfs if needed), so the password never has to
be
    written into an FTP: path. remember = also save it in SD:/etc/ftpfs.ini.
ftpfs_login_site (&site, remember) same, with the port / FTPS / start folder too
(the File Viewer's Connect dialog: they pre-fill the form next time).
ftpfs_forget ("ftp.example.com") drop a login (memory + the file).
ftpfs_load_sites (sites, max) the remembered servers (the Connect dialog's
list).

```

SD:/etc/ftpfs.ini holds one line per server (or .ini sections, see ftpfs_load_sites):

```
host user hexpass [port tls folder]      ("- " = empty user / password)
```

hexpass = the password XOR a fixed key, in hex: OBFUSCATED, NOT ENCRYPTED. IPC: service "ftpfs"; message type 1 = "host\0user\0pass\0[port\0tls\0folder\0]", 2 = the same + remember, 3 = "host\0" forget (see user/BinUtils/ftpfs.cpp).

```

#define FTPFS_SITES_FILE    "SD:/etc/ftpfs.ini"
#define FTPFS_MAXSITES     16

```

```

struct ftpfs_site
{
    char host[128], user[64], pass[64], port[8], folder[128];
    int  tls;           // 1 = FTPS
};

```

```

void ftpfs_obf_hex (const char *in, char *out, int cap); // a password -> its
hexpass (XOR a fixed key, in hex: obfuscated, not encrypted) in out (cap bytes, cut
to fit)

```

```
void ftpfs_unobf_hex (const char *in, char *out, int cap); // a hexpass
(ftpfs_obf_hex) -> the password in out (cap bytes, cut to fit)
```

Parse one file line; 1 if it is a server.

```
int ftpfs_parse_site (const char *line, struct ftpfs_site *s);
```

Format one file line (with '\n'); returns its length.

```
int ftpfs_format_site (const struct ftpfs_site *s, char *out, int cap);
```

A text file saved by a Windows editor: drop a UTF-8 BOM, turn UTF-16 (Notepad) into 8-bit. In place; returns the new length.

```
int ftpfs_text_fix (char *b, int n);
```

The remembered servers, in file order. Returns how many. Also reads the same data written by hand in .ini style:

```
[ftp.example.com]
user = me
password = secret      (plain; or hexpass = <obfuscated>)
port = 21  tls = 1  folder = /www      (one per line, all optional)
```

```
int ftpfs_load_sites (struct ftpfs_site *sites, int max);
int ftpfs_login (const char *host, const char *user, const char *pass, int remember);
int ftpfs_login_site (const struct ftpfs_site *s, int remember);
int ftpfs_forget (const char *host);
```

netkit/http.hpp

http.hpp -- a small reusable HTTP/1.1 client class for Onyx user apps.

Built on the ABI v21 TCP socket kapis (kapi_tcp_connect/send/recv/close + kapi_net_status). Aimed at REST / web-API clients: custom request headers (Authorization, Accept, ...), a request body with a chosen Content-Type, and a parsed response (status code + headers + body, with chunked transfer decoded).

Design (matches the Onyx user model, like httpc.h / uikit.hpp):

- * Header-only, freestanding C++ -- works in EVERY app (integer-only GUI apps and newlib apps alike). It does NOT use new/delete, the STL, or libc; only kapi and its own byte-loop helpers. Compile it into one translation unit (one app).
- * The CALLER provides the response buffer (char buf[N]); the response body/headers point INTO it. No dynamic allocation. `truncated` flags an overflow.
- * Connection: close -- one request per connection; we read until the server closes (or Content-Length / chunked tells us the body is complete), then parse.
- * Blocking: polls the non-blocking kapi_tcp_recv with a timeout (briefly stalls the calling app, like an IRC connect). Fine for occasional API calls.

HTTPS: NetSurf and friends get HTTPS from an external TLS stack (libcurl+OpenSSL); Onyx has no TLS yet. This class RECOGNIZES https:// and returns HTTP_ERR_HTTPS so call sites are already HTTPS-shaped. When mbedTLS lands, only the transport helpers (connect/send/recv) below change -- the public API stays identical.

```
Example (REST):
char buf[32*1024];
HttpClient api;
api.user_agent("myapp/1.0").bearer(token).accept("application/json");
HttpResponse r = api.get("http://api.example.com/v1/items", buf, sizeof buf);
if (r.ok()) { /* parse r.body (NUL-terminated), r.body_len */ }
```

```
HttpResponse p = api.post_json("http://api.example.com/v1/items",
                               "{\"name\":\"x\"}", buf, sizeof buf);
```

Define ONYX_HTTP_TLS (and link the mbedTLS libs) BEFORE including this header to enable https://. Without it the class is fully freestanding and https:// returns HTTP_ERR_HTTPS. The TLS glue (transport) lives in user/Libs/tls/onyx_tls.hpp.

Transport / protocol errors are reported as a NEGATIVE HttpResponse::status.

```
enum HttpError
{
    HTTP_ERR_BAD_URL = -1, // could not parse the URL / empty host
    HTTP_ERR_HTTPS   = -2, // https:// requested but TLS not compiled in (no
ONYX_HTTP_TLS)
    HTTP_ERR_NO_NET   = -3, // the network link is down
    HTTP_ERR_CONNECT  = -4, // DNS or TCP connect failed
    HTTP_ERR_SEND     = -5, // send() failed
    HTTP_ERR_TIMEOUT  = -6, // no response within the timeout
    HTTP_ERR_EMPTY    = -7, // connection closed with no data
    HTTP_ERR_TLS      = -8, // TLS handshake / protocol error
};
```

```
namespace http_detail
{
```

The length of the string s (0 for a null pointer).

```
int slen (const char *s);
```

c in lower case ('A'..'Z' only).

```
char lc (char c);
```

Append s (or n bytes of s) to d[*o], never overflowing cap; keeps d NUL-terminated.

```
void cat (char *d, int cap, int *o, const char *s);
void catn (char *d, int cap, int *o, const char *s, int n);
inline void cati (char *d, int cap, int *o, long v) // non-negative decimal
{
    char t[20]; int n = 0;
}
```

Case-insensitive lookup of a header in [hdr, hdr+hlen). Returns the value start (trimmed) + its length via *vlen, or 0 if absent.

```
const char *find (const char *hdr, int hlen, const char *name, int *vlen);
```

The decimal number at the start of [s, s+n) (its leading digits, no sign) → its value, 0 if there is no digit.

```
int parse_int (const char *s, int n);
```

Case-insensitive: does [s,s+n) contain the word w?

```
bool contains_ci (const char *s, int n, const char *w);
```

Decode HTTP chunked transfer-coding in place over [b, b+len). Returns new length.

```
int dechunk (char *b, int len);
```

Does the string s start with pfx, the case ignored?

```
bool starts_ci (const char *s, const char *pfx);
```

Parse "[http(s)://]host[:port][/path]". Sets *https, *port, host, path("/").

```
bool parse_url (const char *url, char *host, int hcap, unsigned *port, char *path,
int pcap, bool *https);
```

Resolve a redirect Location against the current absolute URL base, into out[cap]. Handles absolute (http(s)://), protocol-relative (//host/...), absolute-path (/...) and path-relative Locations. Returns false on parse error.

```
bool resolve_redirect (const char *base, const char *loc, char *out, int cap);
}
```

transport: plain TCP, or TLS (mbedTLS) when ONYX_HTTP_TLS is defined

A thin send/rcv/close seam so HttpClient is oblivious to http vs https. rcv() keeps the kapi_tcp_rcv convention: >0 bytes, 0 = nothing yet (poll), <0 = closed.

```
struct Transport
{
    int sock; // kapi TCP handle
    bool tls;
#ifdef ONYX_HTTP_TLS
    onyx_tls::Session ssl;
#endif
};
```

```
namespace http_detail
{
```

Open the connection (and TLS handshake if requested). 0 ok, else an HttpError; on connect failure *raw gets the raw kapi_tcp_connect code (for diagnostics).

```
int tp_open (Transport &t, const char *host, unsigned port, bool tls, int *raw);
int tp_send (Transport &t, const void *b, int n);
inline int tp_rcv (Transport &t, void *b, int n) // >0 / 0 = none yet / <0 =
closed
{
    return kapi_tcp_rcv (t.sock, b, (unsigned) n);
} inline void tp_close (Transport &t);
kapi_tcp_close (t.sock);
}
```

A parsed HTTP response. body/hdr point INTO the caller's buffer; body is NUL-terminated. status is the HTTP code (e.g. 200, 404) or a negative HttpError.

```
struct HttpResponse
{
    int status;
    bool truncated; // the response did not fit in the buffer
    const char *body;
    int body_len;
    const char *hdr; // header block (between the status line and the blank line)
    int hdr_len;
    int net_err; // raw kapi_tcp_connect code when
status==HTTP_ERR_CONNECT
// (-1 no net, -2 too many sockets, -3 DNS fail, -5 connect); else 0
```

```

bool ok (void)          const { return status >= 200 && status < 300; }
bool is_error (void)    const { return status < 0; } // transport/protocol error

// Copy a response header value (case-insensitive) into out[cap], NUL-terminated.
// Returns the value length (may exceed cap-1 if truncated), or -1 if absent.
int header (const char *name, char *out, int cap) const
{
    int vl = 0;
    const char *v = http_detail::find (hdr, hdr_len, name, &vl);
    if (!v) { if (cap > 0) out[0] = '\0'; return -1; }
    int n = vl < cap - 1 ? vl : cap - 1;
    for (int i = 0; i < n; i++) out[i] = v[i];
    if (cap > 0) out[n] = '\0';
    return vl;
}
};

```

Reusable HTTP/1.1 client. Configure default headers once (chainable), then issue any number of requests. Stateless across calls except for the default headers.

```

class HttpClient
{
public:
    HttpClient (void) : m_hdrs_len (0), m_timeout (15000), m_ua ("Onyx/1.0"), m_prog
(0), m_progCtx (0)
    {
        m_hdrs[0] = '\0';
    }

    // --- configuration (chainable) -----
    HttpClient &user_agent (const char *ua) { m_ua = ua ? ua : "Onyx/1.0"; return
*this; }
    HttpClient &timeout_ms (unsigned ms) { m_timeout = ms; return *this; }
    // Told the bytes received so far (headers included) while a response comes in: a
download's
    // progress. fn returns false to stop (the response then ends there, truncated).
    HttpClient &progress (bool (*fn) (void *ctx, long got), void *ctx) { m_prog = fn;
m_progCtx = ctx; return *this; }
    void reset_headers (void) { m_hdrs_len = 0; m_hdrs[0] = '\0'; }

    // Add a default header line ("Name: Value") sent with every request.
    HttpClient &header (const char *name, const char *value)
    {
        using namespace http_detail;
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, name);
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, ": ");
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, value);
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, "\r\n");
        return *this;
    }
    HttpClient &bearer (const char *token) // Authorization: Bearer <token>
    {
        using namespace http_detail;
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, "Authorization: Bearer ");
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, token);
        cat (m_hdrs, (int) sizeof m_hdrs, &m_hdrs_len, "\r\n");
        return *this;
    }
    HttpClient &accept (const char *type) { return header ("Accept", type); }
}

```

```

// --- requests -----
HttpResponse get (const char *url, char *buf, int cap)
{ return send ("GET", url, 0, 0, 0, buf, cap); }
HttpResponse del (const char *url, char *buf, int cap)
{ return send ("DELETE", url, 0, 0, 0, buf, cap); }
HttpResponse post (const char *url, const char *ctype, const void *body, int len,
                  char *buf, int cap)
{ return send ("POST", url, ctype, body, len, buf, cap); }
HttpResponse put (const char *url, const char *ctype, const void *body, int len,
                  char *buf, int cap)
{ return send ("PUT", url, ctype, body, len, buf, cap); }
HttpResponse patch (const char *url, const char *ctype, const void *body, int
len,
                  char *buf, int cap)
{ return send ("PATCH", url, ctype, body, len, buf, cap); }

```

```

// JSON convenience (Content-Type: application/json); `json` is a C string.
HttpResponse post_json (const char *url, const char *json, char *buf, int cap)
{ return send ("POST", url, "application/json", json, http_detail::slen (json),
buf, cap); }
HttpResponse put_json (const char *url, const char *json, char *buf, int cap)
{ return send ("PUT", url, "application/json", json, http_detail::slen (json),
buf, cap); }

```

```

// The general request, following up to 8 HTTP 3xx redirects (resolving relative
// Locations). 301/302/303 switch a non-GET to GET (curl's default); 307/308 keep
// the method + body. Returns the FINAL response (or the last 3xx if it has no
// Location / the hop limit is hit). body/hdr point into buf[cap]; <0 =
HttpError.
HttpResponse send (const char *method, const char *url, const char *ctype,
                  const void *body, int len, char *buf, int cap)
{
    using namespace http_detail;
    char cur[2048]; int o = 0; cur[0] = '\0';
    cat (cur, (int) sizeof cur, &o, url);

```

```

    HttpResponse r = send_once (method, cur, ctype, body, len, buf, cap);
    for (int hop = 0; hop < 8; hop++)
    {
        if (r.status < 300 || r.status >= 400 || r.status == 304)
            return r; /* not a redirect we follow */
        char loc[2048];
        if (r.header ("Location", loc, (int) sizeof loc) <= 0)
            return r; /* 3xx without Location: hand it back */
        char next[2048];
        if (!resolve_redirect (cur, loc, next, (int) sizeof next))
            return r;
        o = 0; cur[0] = '\0';
        cat (cur, (int) sizeof cur, &o, next);
        if (r.status != 307 && r.status != 308)
            { method = "GET"; body = 0; len = 0; ctype = 0; } /* 301/302/303 -> GET
*/
        r = send_once (method, cur, ctype, body, len, buf, cap);
    }
    return r; /* too many redirects */
}

```

```

// One request, no redirect handling. method = "GET"/"POST"/...; ctype/body
optional.
// The response is read into buf[cap]; body/hdr point into it. Negative status =

```

HttpError.

```
HttpResponse send_once (const char *method, const char *url, const char *ctype,
                        const void *body, int len, char *buf, int cap)
{
    using namespace http_detail;

    HttpResponse r;
    r.status = 0; r.truncated = false;
    r.body = buf; r.body_len = 0; r.hdr = buf; r.hdr_len = 0; r.net_err = 0;
    if (cap < 16) { r.status = HTTP_ERR_BAD_URL; return r; }

    char host[160], path[1024]; unsigned port; bool https;
    if (!parse_url (url, host, (int) sizeof host, &port, path, (int) sizeof path,
&https))
    { r.status = HTTP_ERR_BAD_URL; return r; }
#ifdef ONYX_HTTP_TLS
    if (https) { r.status = HTTP_ERR_HTTPS; return r; }
#endif

    char ip[40];
    if (!kapi_net_status (ip, sizeof ip)) { r.status = HTTP_ERR_NO_NET; return r; }
}

    Transport tp;
    { int orc = tp_open (tp, host, port, https, &r.net_err); if (orc != 0)
{ r.status = orc; return r; } }

    // Build the request line + headers.
    char req[2048]; int o = 0; req[0] = '\0';
    cat (req, (int) sizeof req, &o, method);
    cat (req, (int) sizeof req, &o, " ");
    cat (req, (int) sizeof req, &o, path);
    cat (req, (int) sizeof req, &o, " HTTP/1.1\r\nHost: ");
    cat (req, (int) sizeof req, &o, host);
    if (port != 80) { cat (req, (int) sizeof req, &o, ":"); cati (req, (int)
sizeof req, &o, (long) port); }
    cat (req, (int) sizeof req, &o, "\r\nUser-Agent: ");
    cat (req, (int) sizeof req, &o, m_ua);
    cat (req, (int) sizeof req, &o, "\r\nConnection: close\r\n");
    catn (req, (int) sizeof req, &o, m_hdrs, m_hdrs_len);
    if (ctype && ctype[0])
    { cat (req, (int) sizeof req, &o, "Content-Type: "); cat (req, (int) sizeof
req, &o, ctype);
      cat (req, (int) sizeof req, &o, "\r\n"); }
    if (body && len > 0)
    { cat (req, (int) sizeof req, &o, "Content-Length: "); cati (req, (int)
sizeof req, &o, len);
      cat (req, (int) sizeof req, &o, "\r\n"); }
    cat (req, (int) sizeof req, &o, "\r\n");

    if (tp_send (tp, req, o) < 0)
    { tp_close (tp); r.status = HTTP_ERR_SEND; return r; }
    if (body && len > 0 && tp_send (tp, body, len) < 0)
    { tp_close (tp); r.status = HTTP_ERR_SEND; return r; }

    // Read until the server closes, the buffer fills, or we idle past the
timeout.
    int total = 0;
    unsigned start = kapi_get_ticks ();
    for (;;)

```

```

    {
        if (total >= cap - 1) { r.truncated = true; break; }
        int n = tp_recv (tp, buf + total, cap - 1 - total);
        if (n > 0)
        {
            total += n; start = kapi_get_ticks ();
            if (m_prog && !m_prog (m_progCtx, total)) { r.truncated = true;
break; }
        }
        else if (n == 0)
        {
            if ((kapi_get_ticks () - start) * 10 > m_timeout) // (ticks: HZ
= 100)
            {
                if (total == 0)
                { tp_close (tp); r.status = HTTP_ERR_TIMEOUT; return r; }
                break;
            }
            kapi_msleep (5);
        }
        else break; // connection closed -> response complete
    }
    buf[total] = '\0';
    tp_close (tp);
    if (total == 0) { r.status = HTTP_ERR_EMPTY; return r; }

    // Status line: "HTTP/1.x NNN reason".
    int i = 0;
    while (i < total && buf[i] != ' ') i++;
    int code = 0;
    if (i < total) { i++; while (i < total && buf[i] >= '0' && buf[i] <= '9')
{ code = code * 10 + (buf[i] - '0'); i++; } }
    r.status = code;

    // Header block = after the status-line LF, up to the blank line.
    int hs = 0; while (hs < total && buf[hs] != '\n') hs++; if (hs < total) hs++;
    int he = -1;
    for (int k = hs; k + 3 < total; k++)
        if (buf[k] == '\r' && buf[k+1] == '\n' && buf[k+2] == '\r' && buf[k+3] ==
'\n') { he = k; break; }
    if (he < 0) { r.hdr = buf + hs; r.hdr_len = total - hs; r.body = buf + total;
r.body_len = 0; return r; }
    r.hdr = buf + hs; r.hdr_len = he - hs;

    int bstart = he + 4;
    int blen = total - bstart;

    int vl = 0;
    const char *te = find (r.hdr, r.hdr_len, "Transfer-Encoding", &vl);
    if (te && contains_ci (te, vl, "chunked"))
        blen = dechunk (buf + bstart, blen);
    else
    {
        const char *cl = find (r.hdr, r.hdr_len, "Content-Length", &vl);
        if (cl) { int n = parse_int (cl, vl); if (n >= 0 && n < blen) blen = n; }
    }

    r.body = buf + bstart;
    r.body_len = blen;
    buf[bstart + blen] = '\0'; // bstart+blen <= total <= cap-1

```

```

        return r;
    }

private:
    char    m_hdrs[1024]; // concatenated default header lines
    int     m_hdrs_len;
    unsigned m_timeout; // idle timeout in ms
    const char *m_ua;
    bool (*m_prog) (void *, long); // the progress (0: none)
    void *m_progCtx;
};

```